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Quentin Rebergue - Software Engineer

Profile

Looking for a full-time position in software engineering with a focus on low-level systems / hardware integration.

MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded.

Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running , cooking.

Work Experience

- **Software Engineer - Platform team, ARISTA**, Dublin 02/2026 – Now

  Implemented non-zero reset value management for ASIC Physical Table registers across the chip-agnostic Strata infrastructure (SDK parsing, code generation, DMA) in python and C++. Won 1st place at Arista's Hackathon by building an AI agent optimization tool (70% token reduction on internal codebases). Approved for production development.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Implemented 1-on-1 code review mentorship model, improving TA-to-student ratio from 1:30 to 1:20.

School Projects

- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **Linux System Tracing & Storage** – Used strace fault injection and syscall filtering to reverse-engineer a binary. Wrote a Bash script automating Linux storage stacks: loop devices, LUKS2, LVM, dm-linear, dm-delay, and RAID.
- **ARM project** – Built an STM32F429ZI embedded system acting as a UART command server using libgapcom (Protobuf over UART). Implemented a 4-level log subsystem and an MPU6050 gyroscope driver over I2C with FIFO reads.Team of 2.
- **MyHTTPd** – HTTP server in C using epoll with edge-triggered mode and per-client state machine for efficient concurrent connection handling. Includes dynamic buffer management, HTTP request parsing, and response generation.

Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020
- **Scientific Baccalaureate (Engineering & Computer Science) — Honors**, Lycée Chaptal 2016 – 2019

Technical Expertise

- **Programming & Languages:** C, ASM, Shell, VHDL, C++, Bash, C++20, Python, algorithms, data structures.
- **Embedded & Hardware:** ASIC interfacing, ESP32, FPGA, I2C, PCB design, Rasberry Pi, STM32, UART, SPI, RTL.
- **Other** / **DevOps:** GDB, Git, strace, system analysis, valgrind, perf, profiling.
- **Systems:** ASIC interfacing, bootloader, DMA, drivers, FPGA, kernel, Linux, operating systems, real-time, virtual memory.